AIMCR FORENSIC REPORT

AI-Powered Mobile Forensic Intelligence Platform for Autonomous Crime Reconstruction

Case Reference:	Case #2026-089A: Phishing & Wallet Takeover
Case Status:	Active
Date Generated:	2026-07-21 14:01:33
Assigned Investigator:	Lead Investigator John
Evidence Count:	18 artifacts ingested
Forensic Complexity Index:	100/100 (Critical)

Executive Summary

This document compiles explainable investigative findings extracted from mobile device artifacts using the Autonomous Crime Reconstruction and Behavioral Profiling engines. Every finding detailed within this report is traceably backed by a corresponding SHA-256 hashed forensic artifact, maintaining full chain of custody. The contents represent investigative directions and indicators of probability rather than definitive judicial outcomes.

1. Incident Reconstruction

Incident Class: SMS Phishing & Account Takeover

Reconstruction Explanation: A multi-stage intrusion sequence was identified. Initially, a SMS message containing an unverified HTTP link was received. This link was visited using Chrome browser, leading to a file download containing a potentially malicious Android application package (.apk). Subsequently, the user accessed settings and banking applications while receiving security OTP messages, which indicates a highly probable credential harvesting and unauthorized account takeover attempt.

Reconstruction Confidence: 85%

Step	Type	Description	Timestamp	Evidence Refs
1	SMS	Received SMS: 'CRITICAL: Your bank account is suspended. Verify immediately at http://secure-auth-bank.net/login' [Threat Intel Alert: Phishing SMS Sender (High)]	2026-07-21 09:45:43	35544eb9-93ea-45ea-8f35-c937823060a0
2	BrowserHistory	Chrome Browser visited URL: http://secure-auth-bank.net/login [Threat Intel Alert: Phishing Domain (Critical)]	2026-07-21 09:47:13	3971ecb1-b1db-4f42-bfd9-ed6d830fdc02
3	File	File downloaded: bank_update.png (/sdcard/Downloads/bank_update.png) [Threat Intel Alert: bank_update.apk hash (Critical)] [Anti-Forensics: Spoofed Extension Detected]	2026-07-21 09:47:43	e632ff8f-af66-4480-a6e1-49cbdf7e3d5c
4	Application	Application opened: com.android.settings	2026-07-21 09:48:43	4e3db00a-0c43-4184-82a9-997801c1d53f
5	Application	Application opened: com.bank.mobileapp	2026-07-21 09:51:43	b4467b5f-f424-4dcb-bf1c-ce4aa6918eee
6	SMS	Received SMS: 'Your OTP code for transfer of $5,000.00 is 882910. NEVER share this code with anyone.'	2026-07-21 09:52:13	c35dcb5a-8592-431b-8bbc-52f4c4aadac6

Incident Class: Unusual Sensitive Asset Access

Reconstruction Explanation: A sensitive local crypto-wallet application was opened during atypical hours (late-night), preceded by settings adjustments. This possible unauthorized local file manipulation suggests potential device exploitation or unauthorized physical access while the device owner was inactive.

Reconstruction Confidence: 75%

Step	Type	Description	Timestamp	Evidence Refs
1	App Anomaly	Sensitive app access: 'com.crypto.wallet' opened during typical sleep hours. Anomaly score: 0.71	2026-07-21 13:15:43	921ca418-1156-41c3-b410-fa6ec07687d4

2. Behavioral Anomaly Log

The Isolation Forest engine analyzed the user's historical feature vectors (active hours, GPS coordinates, application categories, file extensions) and highlighted the following deviations from baseline behavior:

Timestamp	Type	Score	Anomaly Explanation / Context	Ref
2026-07-21 11:46:43	GPS	1.00	Flagged location anomaly at coordinates (34.0522, -118.2437). This location is far outside the user's standard historical geographic cluster.	789fb47b
2026-07-21 09:35:43	GPS	0.83	Flagged location anomaly at coordinates (40.7128, -74.0060). This location is far outside the user's standard historical geographic cluster.	7994fbef
2026-07-21 11:45:43	GPS	0.77	Flagged location anomaly at coordinates (40.7130, -74.0058). This location is far outside the user's standard historical geographic cluster.	3be520d8
2026-07-21 13:15:43	Application	0.71	High-risk application 'com.crypto.wallet' was accessed at 01:15 PM, deviating from the typical daily routine.	921ca418

2a. Anti-Forensics & Tampering Log

Heuristic scanners inspected the integrity of the device artifacts to find evidence of active concealment, extension spoofing, file system timestamp manipulation, or database tampering:

Anomaly Type	Severity	Description	Evidence Ref
Eraser Application Launched	Medium	Suspicious anti-forensics application launched: CCleaner (Cache/Artifact Eraser) (com.piriform.ccleaner).	26ab4b81...
Extension Spoofing	High	File 'bank_update.png' has a 'png' extension but contains signatures matching an Android application package (.apk).	e632ff8f...
Time-Stomping Detected	High	File 'confidential_transfers.xlsx' modification timestamp is set to UNIX Epoch 0 (1970-01-01), indicating deliberate metadata deletion.	bae57e74...
SQLite DB Tampering	High	SQLite database file 'incognito_history.db' was found with 0 bytes or empty tables, indicating potential table purging or record wiping.	dec22f55...

3. Relationship Network & Priority Entities

The communication graph maps contacts, browser domains, and apps into connected modular communities. Nodes are assigned an Investigative Priority score using degree and betweenness centralities.

Priority Investigation Targets

Entity Identifier	Node Class	Investigative Priority Score	Action Needed
Unknown	Contact	85.0/100	Immediate Forensics Review
secure-auth-bank.net	Location	85.0/100	Immediate Forensics Review
com.crypto.wallet	Application	85.0/100	Immediate Forensics Review
SMS Messenger	Application	21.54/100	Monitor & Validate
Office Partner	Contact	10.0/100	Monitor & Validate
Mom	Contact	10.0/100	Monitor & Validate

3a. Forensic Investigation Recommendations

AI/Deterministic Recommendations Summary: Based on the analysis of active threat intelligence matches and anti-forensics indicators, the investigator should prioritize disassembling the spoofed APK binary (disguised as a PNG) and carving the tampered SQLite database. Reconstructing network infrastructure and verifying modification timestamps via system logs is critical to confirm the attack vector and trace perpetrator activity.

Recommended Actions	Priority	Forensic Rationale / Next Steps
Request cellular routing trace for MSISDN '+18005550199'	High	Flagged by Threat Intel as 'Phishing SMS Sender'. Tracing cell tower connections will identify geographical origin of the sender.
Perform WHOIS and passive DNS resolution query on 'secure-auth-bank.net'	High	The phishing infrastructure domain/URL 'secure-auth-bank.net' was matched. This will identify domain registrars, hosting IPs, and subdomains.
Check proxy/DNS resolution logs on client device for contacts with 'secure-auth-bank.net'	Medium	Determines exact duration and timeline of client communication with the malicious infrastructure.
Extract file matching hash '5a1b3c5e8d7f...' for static disassembly	Critical	The malicious package was detected. JADX/Apktool static disassembly is recommended to map internal Java classes and intent handlers.
Audit permissions and isolation state of package 'com.crypto.wallet'	High	This application was launched during the attack. Confirm if it holds Device Admin, SMS reading, or accessibility permissions.
Examine device cache logs and temp file locations	Medium	Forensic cleaners/cache erasers attempt to delete temporary logs. Recovery of overwritten sectors or unallocated database pages may be required.
Perform file header magic signature verification on spoofed files	Critical	Extension mismatch detected. Run Linux 'file' command or read binary headers directly to confirm MIME structure.

Analyze file system journal records ($UsnJrnl / ext4 journal)	High	Active metadata manipulation (modified time set to Epoch 0 or preceding creation time) requires low-level journal audits to rebuild accurate timelines.
Initiate SQLite database cell/carve recovery on empty database	High	Zero tables or zero size in DB suggests table dropping or vacuuming. Attempt cell carving in unallocated DB space to recover deleted rows.

4. Evidence Integrity Index (Chain of Custody)

Every artifact imported into the platform is indexed below with its original acquisition timestamp and a SHA-256 integrity hash to ensure transparency and auditability.

Evidence ID	Timestamp	Type	Source File	SHA-256 Hash Signature
42acdbcb...	2026-07-16 09:45:43	Contact	Contacts.db	5a1b3c5e8d7f2a1b...
bba47696...	2026-07-16 09:45:43	Contact	Contacts.db	5a1b3c5e8d7f2a1b...
0a2cf194...	2026-07-21 08:15:43	SMS	SMS_Backup_2026.xml	5a1b3c5e8d7f2a1b...
ae4b5702...	2026-07-21 08:45:43	SMS	SMS_Backup_2026.xml	5a1b3c5e8d7f2a1b...
26ab4b81...	2026-07-21 08:45:43	Application	App_Usage_Stats.db	5a1b3c5e8d7f2a1b...
7994fbef...	2026-07-21 09:35:43	GPS	Location_History.db	5a1b3c5e8d7f2a1b...
35544eb9...	2026-07-21 09:45:43	SMS	SMS_Backup_2026.xml	5a1b3c5e8d7f2a1b...
3971ecb1...	2026-07-21 09:47:13	BrowserHistory	Chrome_History.db	5a1b3c5e8d7f2a1b...
e632ff8f...	2026-07-21 09:47:43	File	Downloads_Index.db	5a1b3c5e8d7f2a1b...
4e3db00a...	2026-07-21 09:48:43	Application	App_Usage_Stats.db	5a1b3c5e8d7f2a1b...
b4467b5f...	2026-07-21 09:51:43	Application	App_Usage_Stats.db	5a1b3c5e8d7f2a1b...
c35dcb5a...	2026-07-21 09:52:13	SMS	SMS_Backup_2026.xml	5a1b3c5e8d7f2a1b...
3be520d8...	2026-07-21 11:45:43	GPS	Location_History.db	5a1b3c5e8d7f2a1b...
789fb47b...	2026-07-21 11:46:43	GPS	Location_History.db	5a1b3c5e8d7f2a1b...
921ca418...	2026-07-21 13:15:43	Application	App_Usage_Stats.db	5a1b3c5e8d7f2a1b...
bdba52b9...	2026-07-21 13:17:43	File	Downloads_Index.db	5a1b3c5e8d7f2a1b...
bae57e74...	2026-07-21 13:20:43	File	FileSystem.db	5a1b3c5e8d7f2a1b...
dec22f55...	2026-07-21 13:25:43	File	FileSystem.db	5a1b3c5e8d7f2a1b...

5. System Investigation Audit Trail

Immutable audit log records detailing investigator queries, report configurations, and analyses run inside this case file:

Timestamp	Action	Investigator details / Queries

2026-07-21 08:29:32	RUN_ANALYSIS	Executed AI behavioral profiling & communication graph analysis for case ID: 2c045cb8-845b-4d0c-b22f-ab6b4aadf10e. Detected 4 anomalies and 4 Threat Intel match(es).
2026-07-21 08:28:21	LOGIN	User investigator@aimcr.gov logged in successfully